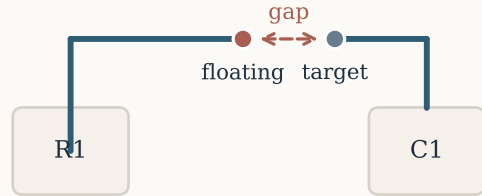


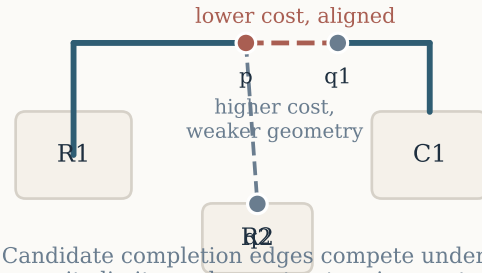
— existing detected wire    - - candidate / recovered completion

Base join leaves a floating pin



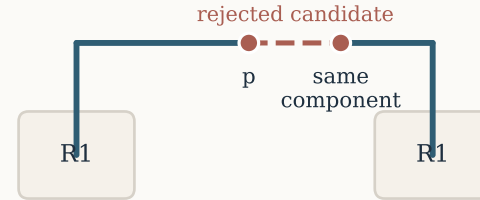
Conservative endpoint-graph joining avoids false shorts but some terminals remain disconnected.

Degree-budget completion scores candidates



Candidate completion edges compete under distance-based capacity limits, and an opt-out assignment.

Self-loop guard rejects the wrong edge



A completion that would collapse two pins from the same component into one net is rejected before merging.

Guarded merge restores the intended net



Only merges that satisfy degree budgets and self-loop guards are added to the recovered net structure.